Supplemental Notes II: Clarifications from J Charles

Nov 2, 2026 · @Shahriar Akhavan Tafti

Follow-up to Supplemental Notes for Further Data Analysis. J Charles confirmed the power-analysis design over text; meeting on campus Nov 2, 2026 after 2 pm to finish the analysis.

Resolved

Question	Answer from J Charles
What is the "20 x 16 x 8 x 2" in the notebook?	Not a real design. It should just be 2 x 3 x 5 (session x period x region). Drop it.
Is region a factor in the power analysis?	Yes, in the main models (2 x 3 x 5 and 2 x 3 x 5 x F). Per-region runs are an extra tier on top, not a replacement.
Why add per-region models?	Concern that variance gets diluted across many factors and statistical power is too low for the full models.
Theta and gamma together or separate?	Separate. Every model runs once for theta and once for gamma.
Band-level DV	Power averaged across the whole band.
Written methods for the last run	Requested so J Charles can describe the analysis accurately. Agreed to provide.

The interpretation floated over text (16 = cue pairs, 8 = rats, 2 = bilateral regions or sessions) is moot now that the notation is dropped, but 8 rats is worth confirming as the n.

Final analysis design

DV: theta or gamma power per region. Windows: 10 s each of baseline, Cue 1, Cue 2, on Day 1 and Day 4. Each model runs once for theta and once for gamma.

Tier	Design	DV	Models
1	2 (session) x 3 (period: baseline, cue 1, cue 2) x 5 (region)	Band-averaged power	2
2	2 (session) x 3 (period) x 5 (region) x F (frequency bin)	Power per bin	2
3a	2 (session) x 3 (period), each region separately	Band-averaged power	10
3b	2 (session) x 3 (period) x F, each region separately	Power per bin	10

Total: 24 models.

To settle at today's meeting

- [ ] Confirm n = 8 rats
- [ ] Power measure and y-axis unit (μV²/Hz, dB, μV, or % change/z if normalized)
- [ ] Per-rat normalization method
- [ ] Multiple-comparison correction across the 10 per-region models in each of tiers 3a and 3b
- [ ] GEE reference category for frequency (5 or 12 Hz?)
- [ ] Figure layout per region (6: 3 theta, 3 gamma)
- [ ] Scope of Granger directionality and theta-gamma coupling on the significant coherence results
- [ ] Per-stimulus gamma coding (side question)
- [ ] Get the example-graph sheet
- [ ] Draft the written methods description of the last run